Student Performance Prediction System

Project Id: AIML-5

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1. Problem Description

The Air Quality Prediction System is a machine learning-based application developed to predict air quality levels using pollutant concentration data. With increasing pollution levels, it is essential to monitor and predict air quality for public health and environmental planning. This system automates the process of analyzing pollutant data such as PM2.5, PM10, NO_2, SO_2, CO, and O_3 and classifies air quality into categories like Good, Satisfactory, Moderate, Poor, Very Poor, and Severe.

The project uses machine learning techniques including data preprocessing, feature selection, classification algorithms, and model evaluation to accurately predict AQI categories. This reduces manual analysis and provides a scalable solution for real-time air quality monitoring.

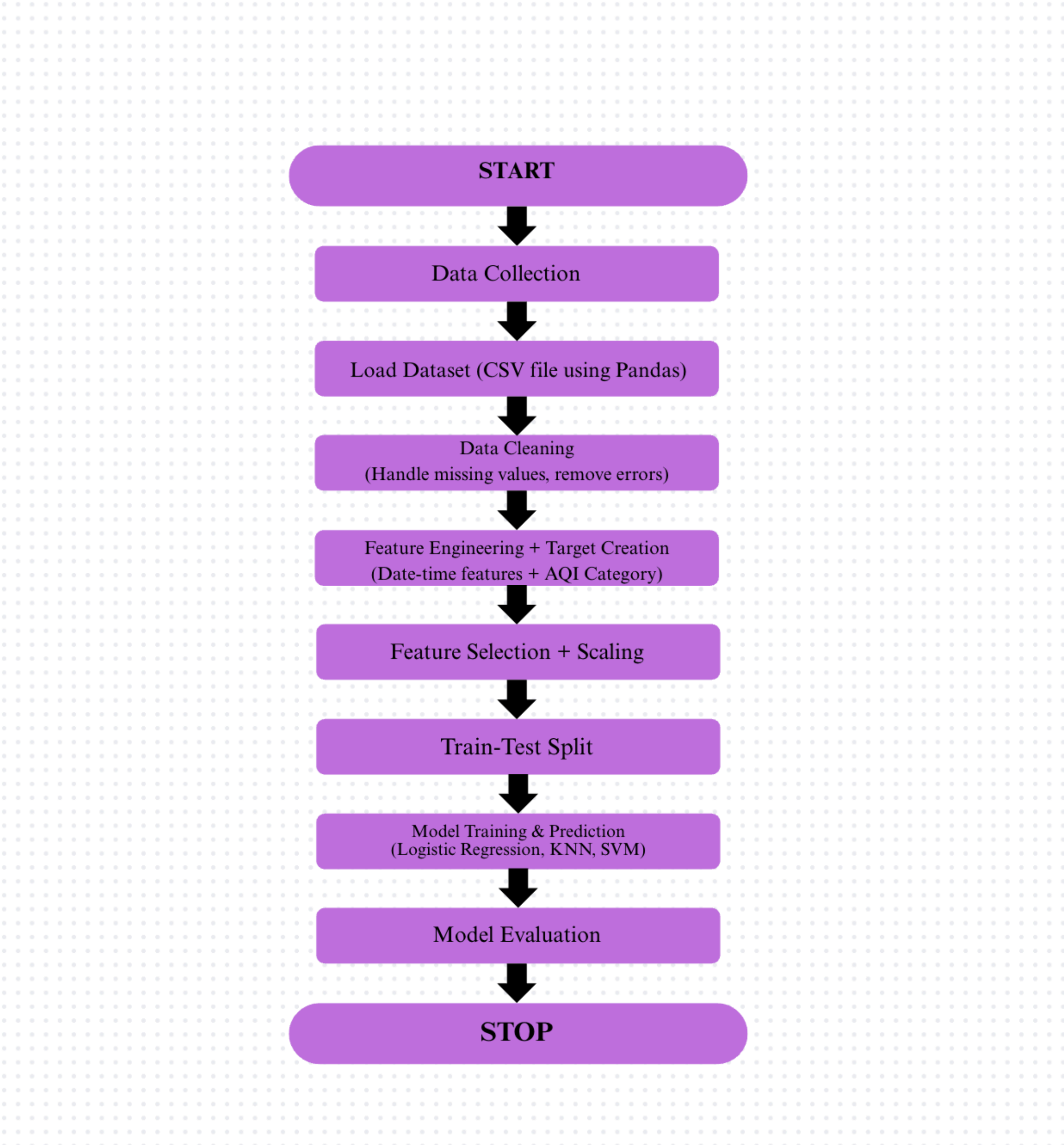
1.1 Dataset Description

The dataset used in this project contains air pollutant concentration data collected over time. The features include:

- CO (Carbon Monoxide)
- NO (Nitric Oxide)
- NO_2 (Nitrogen Dioxide)
- O_3 (Ozone)
- SO_2 (Sulfur Dioxide)
- PM2.5 (Fine Particulate Matter)
- PM10 (Coarse Particulate Matter)
- NH_3 (Ammonia)
- Date and Time

Since the dataset does not include AQI categories, they are generated based on PM2.5 concentration using standard AQI thresholds.

1.2 Flow Diagram



2. Modules for Project Implementation

2.1 Modules Used

1. Pandas

Pandas is used for data manipulation and analysis. It helps in loading the dataset using read_csv(), viewing initial data with head(), and handling missing values using fillna(). It plays an important role in organizing and preparing the dataset for further processing.

2. NumPy

NumPy is used for numerical operations and efficient handling of arrays. It supports mathematical computations required during preprocessing and feature transformation.

3. Matplotlib and Seaborn

Matplotlib and Seaborn are used for data visualization. They help in plotting graphs and heatmaps to understand relationships between different pollutant features and analyze patterns in the dataset. Visualization also helps in interpreting model performance.

4. Scikit-learn

Scikit-learn is used for machine learning tasks. StandardScaler() is applied to normalize the data, train_test_split() is used to divide the dataset into training and testing sets, and SelectKBest() is used for feature selection. Classification models such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) are used to predict air quality categories.

Conclusion

These modules collectively help in data preprocessing, feature selection, model training, and evaluation, forming a complete machine learning pipeline for air quality prediction.

2.2 Platform Used:

Jupyter Notebook – For implementation, visualization, and testing

Python – Programming language used for development

3. Machine Learning Model/ Searching Technique Used

1. Logistic Regression

Logistic Regression is a supervised learning algorithm used for classification problems. It predicts the probability of a data point belonging to a particular class based on input features and assigns it to the most likely category. It works well for linearly separable data and is simple and efficient.

2. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a distance-based classification algorithm. It classifies a data point based on the majority class of its ‘k’ nearest neighbors. The similarity between data points is usually measured using distance metrics like Euclidean distance. It is simple and effective for small datasets.

3. Support Vector Machine (SVM)

Support Vector Machine is a powerful classification algorithm that finds the optimal hyperplane to separate different classes. It works well for both linear and non-linear data by using kernel functions. SVM is effective in high-dimensional spaces and provides good accuracy.

4. Evaluation metrics used

1. Accuracy

Accuracy measures the overall correctness of the model. It is defined as the ratio of correctly predicted observations to the total observations.

Formula:

Accuracy = (Correct Predictions / Total Predictions)

2. Precision

Precision measures how many of the predicted positive cases are actually correct. It is important when the cost of false positives is high.

Formula:

Precision = TP / (TP + FP)

3. Recall

Recall measures how many of the actual positive cases are correctly identified by the model. It is useful when missing a positive case is critical.

Formula:

Recall = TP / (TP + FN)

4. F1 Score

F1 Score is the harmonic mean of precision and recall. It provides a balance between both metrics, especially useful for imbalanced datasets.

Formula:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

5. Confusion Matrix

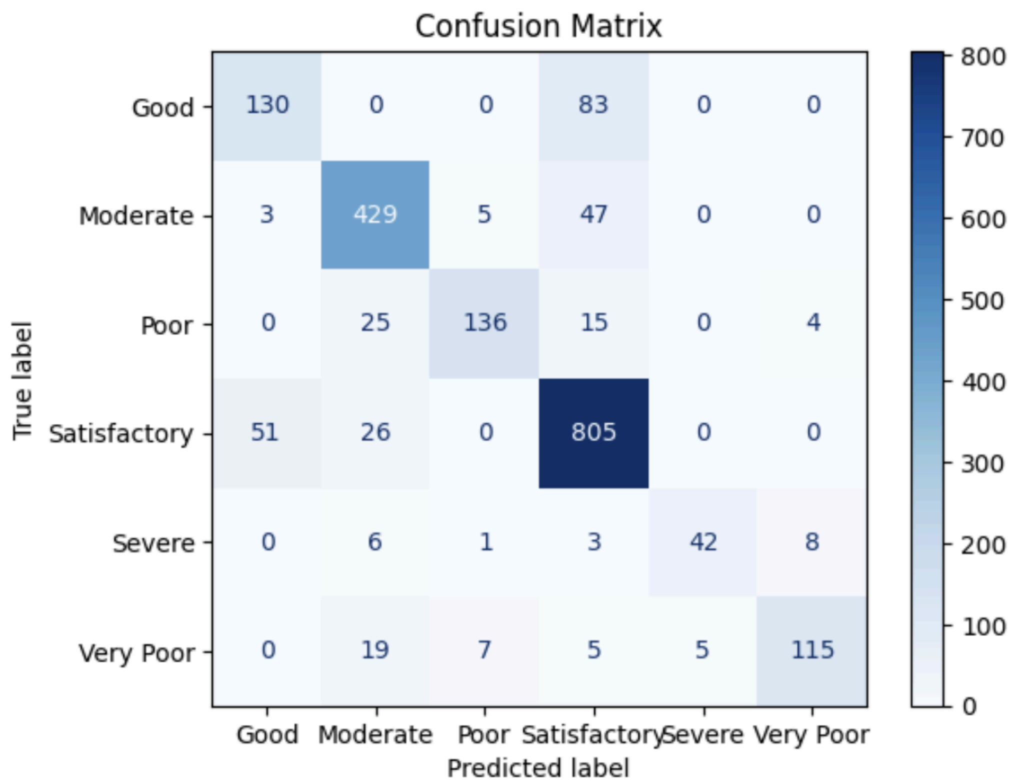
A Confusion Matrix is a table used to evaluate the performance of a classification model. It shows the number of correct and incorrect predictions categorized as True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

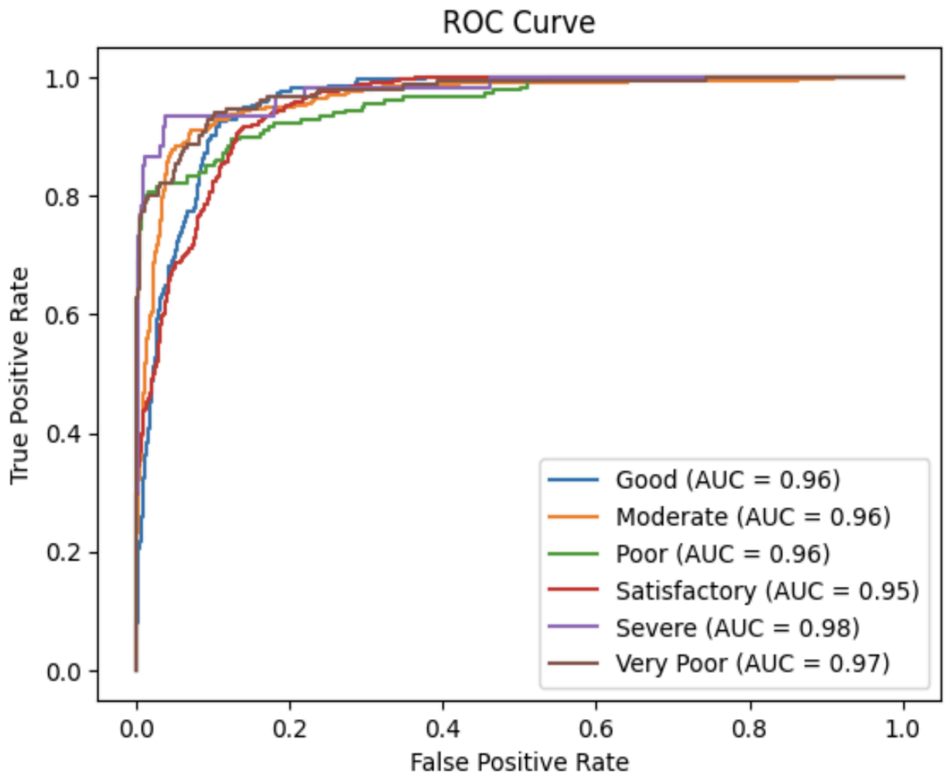
6. ROC Curve

The Receiver Operating Characteristic (ROC) Curve is a graphical representation of a model's performance. It plots the True Positive Rate (Recall) against the False Positive Rate, helping to evaluate the model's ability to distinguish between classes.

5. Results and Visualizations

- Correlation heatmap between pollutant features
- Model comparison graph (accuracy of LR, KNN, SVM)
- Confusion matrix visualization
- Feature importance analysis





Classification Report:

	precision	recall	f1-score	support
Good	0.71	0.61	0.65	213
Moderate	0.85	0.89	0.87	484
Poor	0.91	0.76	0.83	180
Satisfactory	0.84	0.91	0.88	882
Severe	0.89	0.70	0.79	60
Very Poor	0.91	0.76	0.83	151
accuracy			0.84	1970
macro avg	0.85	0.77	0.81	1970
weighted avg	0.84	0.84	0.84	1970

6. Conclusion and Future Scope:

This project successfully predicts air quality categories using machine learning models. Among the implemented models, the best-performing model provides accurate classification of AQI levels.

Future Scope:

- Integration with real-time sensor data
- Use of advanced models like Random Forest or Deep Learning
- Development of a web or mobile application
- Inclusion of weather parameters for better prediction

7. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

• GeeksforGeeks – https://www.geeksforgeeks.org/

• TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm

• Stack Overflow – https://stackoverflow.com